

## SANTA BARBARA 11 W, CA, USA

WMO: 745050

Lat: 34.4141N

Lon: 119.8796W

Elev: 6

StdP: 101.26

Time Zone: -8.00 (NAP)

Period: 08-19

WBAN: 53152

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | 2.9        | 4.1 | -4.6                          | 2.6 | 10.5 | -2.9 | 3.0 | 9.7  | 6.5                   | 14.4 | 5.7 | 14.1 | 1.0                   | N/A  | 0.371 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 9             | 8.3                    | 26.4            | 18.4 | 24.7 | 18.3 | 23.4 | 18.0 | 20.8                | 24.5 | 19.9 | 23.4 | 19.2 | 22.5 | 2.4                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.4                            | 14.2 | 23.3 | 18.4 | 13.3 | 22.2 | 17.7 | 12.7 | 21.5 | 60.1          | 24.3 | 56.8 | 23.5 | 54.5 | 22.6 | 23.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 5.7               | 4.8  | 3.9 | 0.8                        | 31.3 | 1.2                | 2.5 | -0.1                                               | 33.1 | -0.8       | 34.5 | -1.4       | 35.9 | -2.3       | 37.7 |    |    |
|                   |      |     | -1.4                       | 21.7 | 1.0                | 0.9 | -2.1                                               | 22.3 | -2.7       | 22.8 | -3.2       | 23.3 | -3.9       | 23.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 14.9   | 12.0  | 11.8  | 13.0  | 14.0  | 15.1  | 16.3  | 18.0  | 18.1  | 18.2  | 16.8  | 14.0  | 11.6 |
|                                                                                  | DBStd       | 3.17   | 2.24  | 2.18  | 2.24  | 2.03  | 1.68  | 1.31  | 1.77  | 1.80  | 2.21  | 2.36  | 2.16  | 2.24 |
|                                                                                  | HDD10.0     | 26     | 8     | 6     | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 9    |
|                                                                                  | HDD18.3     | 1326   | 198   | 182   | 167   | 131   | 104   | 63    | 27    | 26    | 27    | 60    | 131   | 209  |
|                                                                                  | CDD10.0     | 1823   | 68    | 57    | 93    | 121   | 158   | 189   | 248   | 252   | 247   | 211   | 122   | 58   |
|                                                                                  | CDD18.3     | 79     | 0     | 0     | 1     | 1     | 3     | 2     | 17    | 19    | 23    | 13    | 1     | 0    |
|                                                                                  | CDH23.3     | 323    | 2     | 2     | 7     | 10    | 22    | 7     | 34    | 41    | 94    | 89    | 15    | 1    |
|                                                                                  | CDH26.7     | 52     | 0     | 0     | 1     | 1     | 4     | 0     | 8     | 2     | 20    | 14    | 2     | 0    |
| Wind                                                                             | WSAvg       | 1.7    | 1.5   | 1.7   | 1.8   | 1.9   | 1.9   | 1.7   | 1.7   | 1.6   | 1.5   | 1.5   | 1.5   | 1.6  |
| Precipitation                                                                    | PrecAvg     | 406    | 82    | 95    | 80    | 24    | 5     | 1     | 1     | 2     | 3     | 13    | 50    | 57   |
|                                                                                  | PrecMax     | 892    | 311   | 346   | 291   | 146   | 46    | 12    | 10    | 48    | 32    | 61    | 176   | 147  |
|                                                                                  | PrecMin     | 110    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | PrecStd     | 171    | 78    | 85    | 72    | 36    | 11    | 3     | 2     | 8     | 7     | 15    | 51    | 48   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 23.0  | 23.1  | 23.8  | 24.7  | 26.6  | 24.3  | 26.9  | 26.3  | 29.8  | 28.7  | 25.3  | 21.4 |
|                                                                                  |             | MCWB   | 11.8  | 13.6  | 14.4  | 13.3  | 14.1  | 16.8  | 20.3  | 20.5  | 17.1  | 15.9  | 15.3  | 13.6 |
|                                                                                  | 2%          | DB     | 20.5  | 20.0  | 20.6  | 21.4  | 21.9  | 21.7  | 23.7  | 24.4  | 25.7  | 25.7  | 22.7  | 19.5 |
|                                                                                  |             | MCWB   | 12.3  | 13.4  | 14.3  | 13.9  | 14.9  | 17.0  | 19.1  | 19.8  | 20.2  | 17.4  | 15.2  | 13.9 |
|                                                                                  | 5%          | DB     | 18.7  | 18.0  | 18.9  | 19.8  | 20.1  | 20.7  | 22.6  | 23.1  | 23.9  | 23.7  | 20.8  | 18.0 |
|                                                                                  |             | MCWB   | 12.8  | 13.4  | 14.0  | 14.0  | 14.9  | 16.7  | 18.7  | 19.0  | 19.1  | 17.5  | 14.7  | 13.4 |
|                                                                                  | 10%         | DB     | 17.2  | 16.6  | 17.5  | 18.4  | 18.9  | 19.9  | 21.5  | 22.0  | 22.6  | 22.2  | 19.6  | 16.8 |
|                                                                                  |             | MCWB   | 13.0  | 13.0  | 13.5  | 13.5  | 14.6  | 16.2  | 18.0  | 18.3  | 18.5  | 17.3  | 14.8  | 13.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 15.7  | 16.2  | 16.9  | 16.7  | 16.8  | 18.6  | 20.9  | 21.3  | 22.0  | 21.5  | 17.9  | 17.4 |
|                                                                                  |             | MCDB   | 18.1  | 18.3  | 21.0  | 20.6  | 20.9  | 22.2  | 25.4  | 25.6  | 25.7  | 24.4  | 21.5  | 19.2 |
|                                                                                  | 2%          | WB     | 15.0  | 15.2  | 15.7  | 15.7  | 16.0  | 17.8  | 19.8  | 20.1  | 21.0  | 20.2  | 16.8  | 15.8 |
|                                                                                  |             | MCDB   | 17.0  | 17.6  | 18.9  | 19.1  | 19.6  | 21.2  | 23.4  | 24.1  | 24.7  | 23.2  | 20.2  | 17.9 |
|                                                                                  | 5%          | WB     | 14.4  | 14.6  | 15.0  | 15.0  | 15.5  | 17.2  | 19.0  | 19.4  | 20.0  | 18.9  | 16.1  | 14.9 |
|                                                                                  |             | MCDB   | 16.4  | 16.9  | 17.8  | 18.2  | 18.8  | 20.4  | 22.4  | 23.0  | 23.3  | 22.4  | 19.4  | 17.0 |
|                                                                                  | 10%         | WB     | 13.8  | 13.9  | 14.2  | 14.4  | 15.0  | 16.6  | 18.4  | 18.7  | 19.2  | 17.9  | 15.4  | 14.0 |
|                                                                                  |             | MCDB   | 15.9  | 16.1  | 16.5  | 17.3  | 18.1  | 19.5  | 21.5  | 21.9  | 22.2  | 21.4  | 18.6  | 16.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.5  | 9.5   | 8.6   | 8.6   | 8.1   | 7.0   | 6.9   | 7.2   | 8.3   | 10.2  | 11.0  | 10.3 |
|                                                                                  |             | MCDBR  | 13.5  | 12.3  | 11.4  | 11.6  | 10.6  | 8.9   | 8.0   | 8.5   | 10.3  | 13.1  | 13.6  | 12.3 |
|                                                                                  |             | MCWBR  | 8.8   | 8.2   | 7.2   | 6.6   | 5.9   | 5.1   | 4.2   | 4.7   | 5.8   | 7.9   | 9.4   | 9.0  |
|                                                                                  | 5% WB       | MCDBR  | 9.7   | 9.6   | 9.2   | 9.3   | 8.6   | 8.0   | 7.7   | 8.3   | 8.9   | 10.9  | 12.0  | 9.7  |
|                                                                                  |             | MCWBR  | 7.3   | 6.9   | 6.4   | 6.1   | 5.2   | 4.6   | 4.1   | 4.6   | 5.1   | 7.2   | 8.8   | 7.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.304  | 0.317 | 0.326 | 0.362 | 0.378 | 0.364 | 0.393 | 0.372 | 0.362 | 0.350 | 0.308 | 0.311 |      |
|                                                                                  | taud        | 2.595  | 2.569 | 2.545 | 2.425 | 2.402 | 2.444 | 2.415 | 2.441 | 2.479 | 2.504 | 2.601 | 2.588 |      |
|                                                                                  | Ebn at Noon | 915    | 936   | 951   | 923   | 908   | 916   | 886   | 903   | 899   | 884   | 901   | 884   |      |
|                                                                                  | Edh at Noon | 77     | 87    | 97    | 115   | 119   | 114   | 116   | 111   | 102   | 92    | 76    | 74    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.07   | 3.94  | 5.31  | 6.62  | 7.29  | 7.56  | 7.30  | 6.83  | 5.86  | 4.50  | 3.46  | 2.78  |      |
|                                                                                  | RadStd      | 0.23   | 0.41  | 0.34  | 0.38  | 0.48  | 0.35  | 0.29  | 0.27  | 0.25  | 0.33  | 0.19  | 0.22  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (37 neighbors) | +0.61 | N/A     | -1.62  | +0.68   | N/A   | N/A   | N/A         | -127    | +220    | +83     |

Nomenclature: See separate page